

Research Paper

A complete control structure based backstepping controller design for stacked multi-cell multi-level SPWM VSC STATCOM

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ABSTRACT

Currently, the installation of FACTS (Flexible Alternative Current Transmission System) devices such as synchronous static compensators (STATCOM) represent a very useful solution to improve both network stability and power quality. The effectiveness of these devices is principally determined by the used static converter (topology and the switching strategy) and the adopted dynamic law characterizing the controller structure. This paper illustrates the stacked five-level, multi-cell STATCOM compensator performances governed by a Backstepping controller technique compared to the use of the conventional PI controller (SPWM) for controlling voltage variation (sags or overshoots) affecting a given network point. The interest in using stacked multi-cell converters topology is to increase the number of voltage levels in order to ensure a good quality voltage waveform at the specified connection point. Therefore, the proposed backstepping controller based on the use of Lyapunov control functions presents a simplified structure and a reduced computational load compared to many others advanced control approaches. The proposed system is verified by running a simulation analysis with Matlab Simulink environment. The simulation results obtained show that the proposed STATCOM configuration ensures good voltage regulation at the common connection point (PCC), correction of the unity power factor on the grid side and good response time via reactive power compensation under variable load conditions.

1. Introduction

Over the last six decades, various high-speed power electronics devices topologies known as FACTS (Flexible Alternative Current Transmission System) intended to be connected to an electrical network have been developed. FACTS slide technology was initiated by EPRI (Electric Power Research Institute) in 1980, which plays an important role in improving the maximum utilization of existing power generation and transmission systems, using power electronic devices (Choudante and Bhole, 2018). FACTS are basically static network controllers that control dynamically the voltage magnitude, impedance and angular phase shift of transmission lines. Thus, to increase energy transfer capacity, reliability, control energy flow, reduce power losses, improve transient stability, dampen power oscillations, improve safety and other system performances (Siddique et al., 2019; Shehata et al., 2021; Choudante and Bhole, 2018).

These controllers are generally classified into three categories depending on how they are connected to the electrical network; in series, in shunt or by a combination of series-series or series-shunt

configurations (Freire et al., 2019; Alajrash et al., 2024; Shehata et al., 2021). These categories these categories are also classified according to two families depending on the technology used. The first one is based on conventional thyristors such as the static var compensator (SVC) and the thyristor controlled series capacitor (TCSC). The second one is based on voltage Source Converters (VSC) such as the static synchronous compensator (STATCOM), the static synchronous series compensator (SSSC), the unified power flow controller (UPFC) and the interline power flow controller (IPFC) etc (Shehata et al., 2022; Sharma et al., 2023).

The STATCOM is one of the most efficient FACTS parallel devices using the voltage source converter (VSC) (Lee and Park, 2019) and its principle of operation is similar to that of the conventional synchronous generator in rotation. It is used to generate or absorb the necessary reactive power in order to ameliorate simultaneously the power factor, voltage regulation and reduce line losses (Gupta and Kumar, 2018; Sadeghi et al., 2011). STATCOM presents several others intrinsic advantages due to its high-power density, small size, minimal losses, fast response and wide compensation range (Ahmed et al., 2021). Conventional two-level, three-phase voltage source converters are used in

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Nomenclature

ANPC	Active Neutral Point Clamped
BSC	Backstepping controller
CHB	Cascade H-bridge
EPRI	Electric Power Research Institute
FACTS	Flexible Alternative Current Transmission System
FC	The flying capacitor
FFT	Fast Fourier Transform
IGBT	insulated-gate bipolar transistor
IPFC	interline power flow controller
MMC	Modular Multilevel Converter
NPC	Neutral point clamped
OCC	one-cycle control
PCC	Point of common connection

PI	Proportional–integral controller
PLL	Phase-locked loop
PS-PWM	Phase-Shifted Pulse Width Modulation
PWM	Pulse width modulation
RMS	Root Mean Square
SMC	Stacked multicell converter
SPWM	Sinusoidal Pulse Width Modulation
SSSC	static synchronous series compensator
STATCOM	Synchronous static compensators
SVC	static var compensator
TCSC	thyristor-controlled series capacitor
THD	Total harmonic distortion
UPFC	unified power flow controller
VSC	Voltage Source Converters

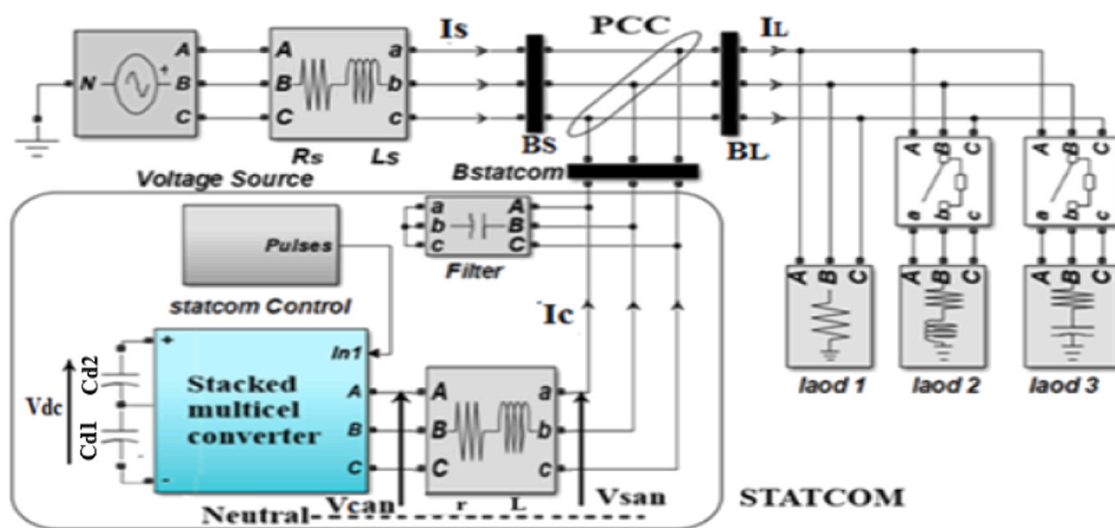


Fig. 1. Schematic representation of a STATCOM connected to the Network.

practice in STATCOM applications. These converters offer advantages such as a reduced number of components, a maximum of two voltage sources and numerous modulation strategies. Disadvantages include poor output voltage quality and high harmonic distortion (Dargahi et al., 2012).

Recently, STATCOMs based on conventional two-level converters have been gradually replaced by multi-level topologies (Kaymanesh et al., 2021); the desired aim being to have reduced switching power losses, improved output voltage quality, a higher number of levels, lower total harmonic distortion (THD) and higher voltage capacity. Moreover, the constraints associated with dangerous variations in switched voltage (dv/dt) are considerably reduced compared with conventional two-level structure. These converters with their different configurations: the neutral point clamped (NPC), the flying capacitor (FC), the cascaded H-bridge (CHB) and the modular multilevel converter (MMC), are generally destined for medium/high power requirements (Daraghi et al., 2014; Ghias et al., 2015; Kamran and Munoz, 2015; Liu et al., 2016).

However, it seems that beyond three voltage output levels, these converters topologies present some disadvantages such as the imbalance of voltages across capacitors and the large number required of clamping diodes relative to NPC topologies, the number of floating capacitors and the imbalance of voltages across these capacitors relative to series multilevel converters, and several isolated DC sources for H-bridge

topologies. These drawbacks have a significant impact on the dimensioning and the price of the converter (Ghias et al., 2015; Shahnian et al., 2015; Baldé et al., 2011).

Over the last two decades, hybrid multilevel converters topologies such as the ANPC (Active neutral point clamped), which is a version derived from the NPC topology, and the SMC (Stacked multicell converter), which is an evolved version of the basic serial multicell topology (with floating capacitors) have been introduced as a competitive alternative for situations requiring less energy storage compared with conventional multilevel topologies (Gateau et al., 2001; Sadigh et al., 2019).

The stacked multicell topology (SMC) that is characterised by its number of cells and stages was proposed in 2001 (Gateau et al., 2001). This topology is capable of generating a higher number of voltage levels and of sharing the voltage stress over several switches. So, the volume of floating capacitors is reduced because the corresponding energy stored is lower compared to the case of series multicell converter (Lienhardt et al., 2005; Premkumar et al., 2017). Consequently, the stacked multicell converter constitute a good choice for multilevel expansion, particularly with a higher switching frequency and high-power density (Lienhardt et al., 2005).

Over the last few decades, several research studies have highlighted the value of integrating STATCOM into the power distribution system using linear control strategies. The work in (Pandey et al., 2017), presents a multilevel cascaded converter based on D-STATCOM using a PI

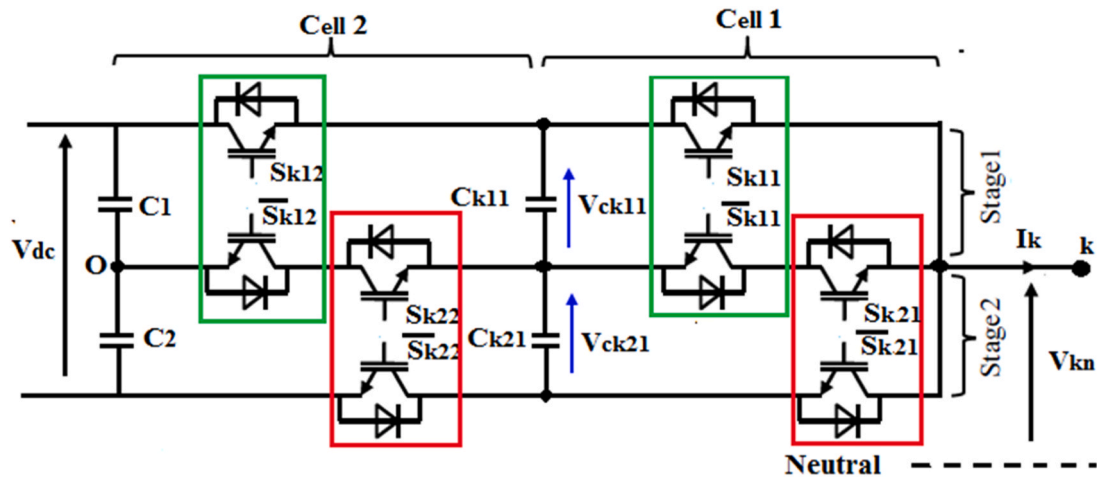


Fig. 2. Circuit diagram of a phase five level of stacked Multicell converter (SMC).

Table 1

States of switches in 4-cell-5-level proposed SM converter.

property	value
The number of the floating capacitors	$n \times p$
The voltages across each floating capacitor	$(i \times V_{dc}) / (n \times p)$, where i represents the rank of the cell ($i: 1, 2, \dots, p - 1$),
The number of possible configurations	$(n + 1)^p$
Number of levels	$(n \times p) + 1$
The apparent frequency of the circuit	$p \times f_{dec}$

controller for reactive energy compensation during grid transients and load changes. In Jeong and Sung (2021) the authors propose a novel one-cycle control (OCC) technique for a static synchronous compensator (STATCOM) based on a five-level clamp diode multilevel converter. A PI control law for a multilevel D-STATCOM CHB is derived to improve the power factor by eliminating harmonics and switching losses is proposed in (Geethamani et al., 2021).

In Zhou and Cheng (2019), the authors propose a new carrier-based semi-polar modulation method for balancing the power loss distribution of the three-level NPC converter operating as a STATCOM. This method is compared to the conventional unipolar modulation method.

In Pan et al. (2022), the authors proposed a new modulated model predictive control (MMPC) scheme for a modular synchronous static compensator based on a cascade converter (STATCOM) (MMCC-STATCOM) in order to compensate the unbalanced load current and regulate the reactive power flow.

In this work, we propose a five-level synchronous multicell stacked static compensator (SMC STATCOM) based on the Backstepping control technique for improving system performance. This control method will enable STATCOMs to supply or absorb the desired reactive power for stabilizing the voltage profile at the common connection point (PCC), so as to avoid fluctuations between source voltage and load voltage and achieve a network power factor close to one. The relatively recent Backstepping is a relatively recent development for non-linear systems. Sequentially and systematically, it enables us to select a Lyapunov function adapted to the system, and to deduce the system control that makes the derivative of this function negative (Mazouz et al., 2021; Roy et al., 2020). It is also capable of replacing the conventional controller due to its simpler structure, the gains of the PI controller are not necessary to determine, harmonics are lower and good robustness to large load variations (Kadivar et al., 2017). To illustrate the advantages of the proposed Backstepping control, a series of simulations were

carried out. The simulation results were analyzed and compared with those obtained using conventional PI control.

2. Description, modelling and control law of the studied STATCOM system

2.1. System dynamics

Fig. 1 shows the structure of the considered STATCOM system. It consists of a DC power source associated to a five-level superimposed multi-cell static converter based on IGBT semiconductors connected in parallel to the mains via a series coupling inductors. The principal task is to regulate the voltage at the common connection point (PCC) by absorbing or supplying reactive energy from the DC bus capacitors. This exchange of reactive energy is ensured by controlling the output voltage magnitudes converter in phase with the mains voltages.

The reactive power transfer between the grid and the STATCOM is done by controlling the voltage DC bus voltage V_{dc} . If the STATCOM output voltage is higher than the mains voltage (PCC), the compensator supplies the reactive power to the mains. If the STATCOM output voltage is lower than the mains voltage (PCC), the compensator absorbs the reactive power. If the two voltages are equal, the reactive power exchanged is zero (Xu and Li, 2014; Mihalic et al., 2016).

In Fig. 1, V_s , V_c and V_L are the source, compensator and load voltages respectively; I_s , I_c and I_L are the source, compensator and load currents respectively. L_s and R_s are the inductance and resistance of the source respectively; L and r are the inductance and internal resistance of the filter coil respectively; V_{dc} , C_{dc1} and C_{dc2} represent the voltage and capacitance of the DC side respectively.

2.2. Converter model

The schematic diagram of one arm of the five-level three-phase superimposed multicell converter is shown in Fig. 2. The modelling of this converter is based on the following adopted assumptions: perfect switches, perfect sources and negligible dead time. The aim of the modelling task is to find a relationship between the control quantities and the electrical quantities of the AC and DC sides of the converter. Each arm of the converter contains two stages and two switching cells separated by two floating capacitors. Each elementary cell of the converter is made up of four bidirectional switches. The outer branches are made up of a single switch and the middle branch is made up of two switches placed in opposite each other (Gateau et al., 2001; Lienhardt et al., 2005; Ghias et al., 2015; Sadigh et al., 2019; Cheng et al., 2021).

The function of the connection of each arm is defined by S_{kij} where i indicates the switch number corresponding to a given cell in the branch

Table 2
States of switches in the proposed 4-cell 5-level SM converter.

Output voltage level V_{ckO}	State of switches				Number of States
	S_{k11}	S_{k12}	S_{k21}	S_{k22}	
$+V_{dc}/2$	1	1	1	1	1
$\frac{V_{dc}}{4}$	0	1	1	1	2
0	1	0	1	1	
	0	0	1	1	1
$-\frac{V_{dc}}{4}$	0	0	0	1	2
	0	0	1	0	
$-V_{dc}/2$	0	0	0	0	1

of phase k ($k = a, b$ or c) of SMC converter, $i = 1, 2, \dots, p$, ($p = 2$) and j defines a particular switch associated with the stage $j = 1, 2, \dots, n$, ($j = 2$). The pairs of the power switch in each branch of the phase (S_{kij} et $\bar{S}_{kij}$) are controlled in a complementary manner. The switch control functions can take two values:

- $S_{kij} = 1$ when the top switch of the branch is closed and the bottom switch is open.
- $S_{kij} = 0$ when the top switch of the branch is open and the bottom switch closed.

The input DC bus is made up of two capacitors in series ($C1$ and $C2$), forming a midpoint denoted (O). Each of these capacitors being regulated to have half the DC bus voltage value ($V_{dc}/2$). So, for a given structure ($n \times p$), we have the following characteristics in (Table 1):

In the particular case corresponding to a SMC 2×2 is traduced by $p = 2$ cells and $n = 2$ stages. It uses $p \times n = 4$ switching cells and $(p - 1) \times n = 2$ floating capacitors.

The state of the commutators of stacked multicell converter with four cells and five levels (SMC 2×2) are illustrated in the [Table 2](#).

Table 2 shows the respective voltages levels according to the converter's configurations in the case where the floating voltages are considered perfectly balanced.

The voltages at the output of the converter with respect to the negative terminal (point O) of the DC bus will be expressed as (Lienhardt et al., 2005):

$$V_{ckO} = (S_{k11} - S_{k12}) \times V_{Ck11} + S_{k12} \times (V_{dc}/2) + (S_{k21} - S_{k22}) \times V_{Ck21} - (1 - S_{k22}) \times (V_{dc}/2) \quad (1)$$

The voltages $V_{ckn} = V_{ckO} - V_{nO}$, ($k = a, b, c$), at the terminals of the load with respect to the neutral point of potential $V_{nO} = (V_{aO} + V_{bO} + V_{cO})/3$, are:

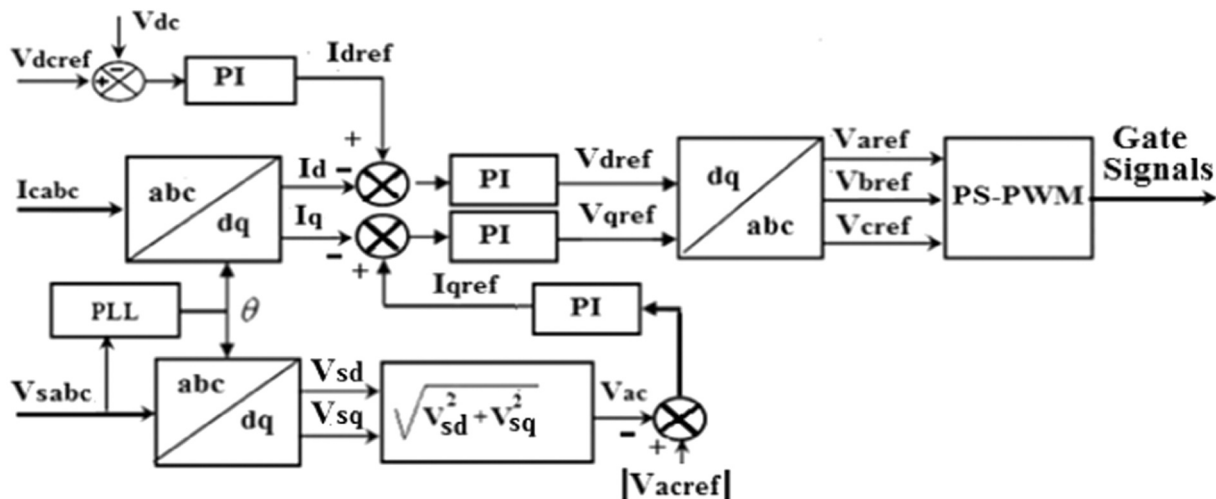


Fig. 3. Block diagram of the STATCOM control strategy based on the PI conventional.

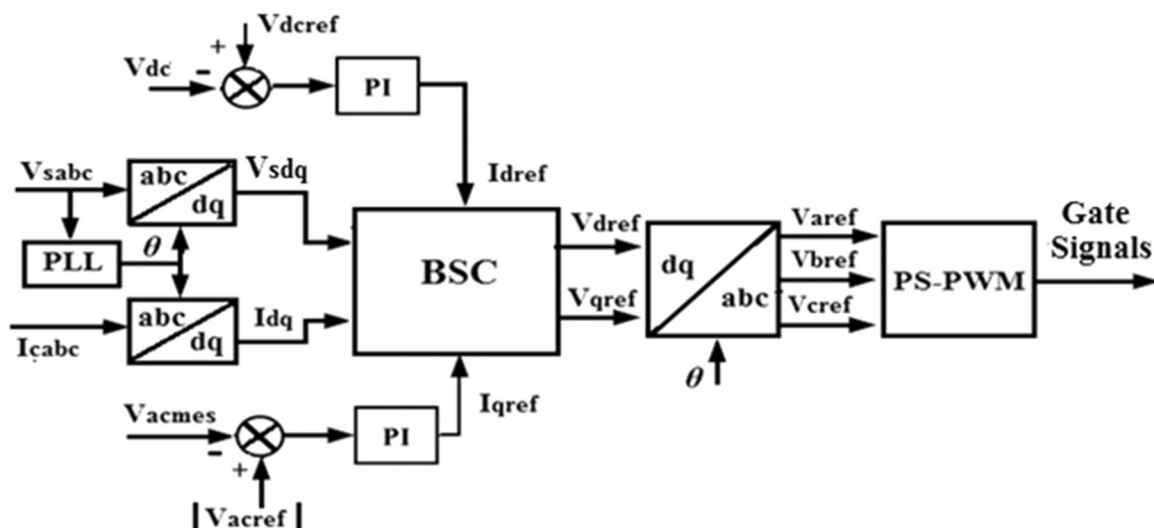


Fig. 4. Block diagram of the STATCOM control strategy based on the Backstepping.

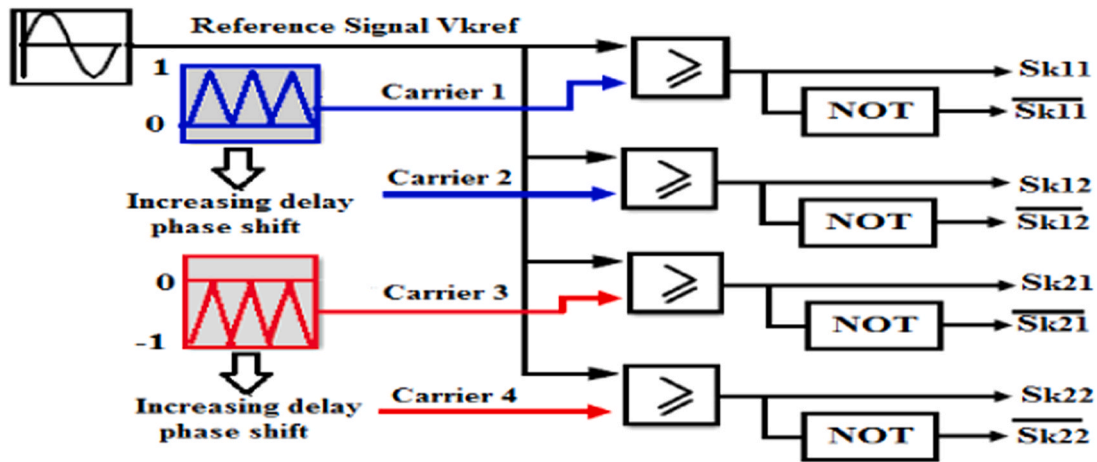


Fig. 5. Simplified diagram of the open loop PWM control of stacked Multi-cell converter.

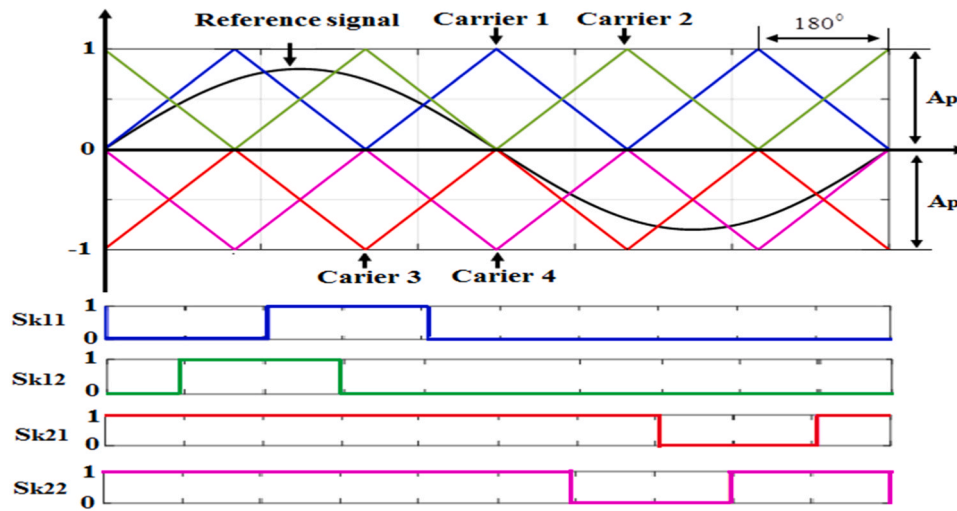


Fig. 6. PS-PWM modulation strategy.

Table 3

Parameters of simulated system (Deniz et al., 2009).

Component	Parameters	Value
AC Source	Line to Line rms Voltage	381 V
	Frequency	50 Hz
	Resistance Rs	7.3 m Ohm
	Inductance Ls	0.23 mH
STATCOM	DC-Link Voltage	750 V
	Inductance Filter L	0.7 mH
Load	Load 1	50 kVA
	Load 2	10 kA/50kVar
	Load 3	10 kA/- 50kVar

$$\begin{bmatrix} V_{can} \\ V_{cbn} \\ V_{ccn} \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \times \begin{bmatrix} V_{ao} \\ V_{bo} \\ V_{co} \end{bmatrix} \quad (2)$$

2.3. STATCOM dynamical model

The STATCOM model is obtained by applying the Kirchhoff's laws to the circuit in Fig. 1, the resulting state model in terms of different electrical variables is formulated as (Freire et al., 2019; Xu and Li, 2014; Murugesan and Muthu, 2011):

$$\begin{cases} V_{san} = rI_{ca} + L \frac{dI_{ca}}{dt} + V_{can} \\ V_{sbn} = rI_{cb} + L \frac{dI_{cb}}{dt} + V_{cbn} \\ V_{scn} = rI_{cc} + L \frac{dI_{cc}}{dt} + V_{ccn} \end{cases} \quad (3)$$

$$\begin{cases} V_{san} = V_m \cos(\omega t) \\ V_{sbn} = V_m \cos(\omega t - (2\pi/3)) \\ V_{scn} = V_m \cos(\omega t - (4\pi/3)) \end{cases} \quad (4)$$

where:

V_{sabc-n} represent the network voltages at the PCC; V_{cabc-n} represent the three-phase voltages at the output of the stacked multicell converter; I_{cabc} represent the three-phase currents circulating from STATCOM; L is the inductance value of the coupling filter and r is the equivalent resistance of the filter.

The Park model corresponding to Eq. (3) is written as:

$$\begin{cases} V_{sd} = rI_d + L \frac{dI_d}{dt} - \omega LI_q + V_d \\ V_{sq} = rI_q + L \frac{dI_q}{dt} + \omega LI_d + V_q \end{cases} \quad (5)$$

In the Eq. (5), ωLI_d and ωLI_q terms represent the cross-coupling terms; ω

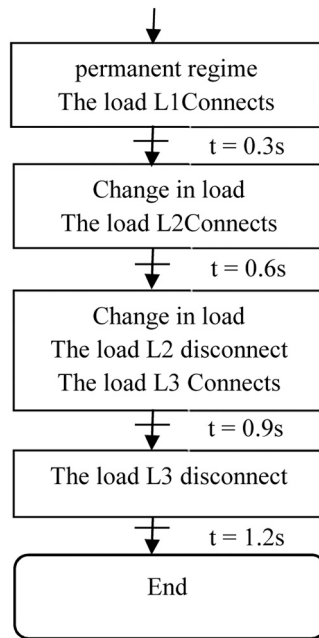


Fig. 7. Script based on STATCOM simulation.

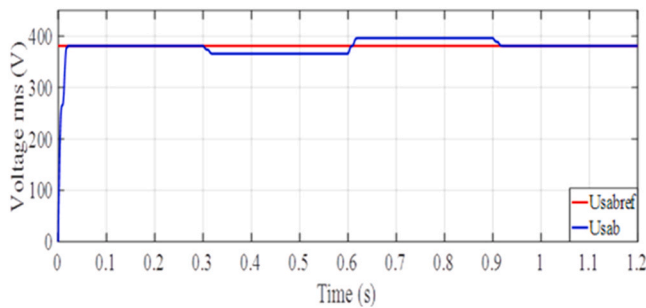


Fig. 8. RMS voltage between phases without STATCOM.

is the grid pulsation; I_d and I_q are d- and q-axis currents of the STATCOM, respectively; V_{sd} and V_{sq} are d- and q-axis voltages of the PCC and V_d and V_q are d- and q-axis, voltages at the output of the stacked multicell converter respectively.

Thus, the expression of the current exchanged with the network is defined by the Eq. (6):

$$\frac{d}{dt} \begin{bmatrix} I_d \\ I_q \end{bmatrix} = \frac{-r}{L} \begin{bmatrix} I_d \\ I_q \end{bmatrix} + \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix} \begin{bmatrix} I_d \\ I_q \end{bmatrix} + \frac{1}{L} \begin{bmatrix} V_{sd} - V_d \\ V_{sq} - V_q \end{bmatrix} \quad (6)$$

The active and reactive powers of the converter are carried out using

the following expressions:

$$P = \frac{3}{2} (V_{sd} I_d + V_{sq} I_q) \quad (7)$$

$$Q = \frac{3}{2} (V_{sd} I_q - V_{sq} I_d) \quad (8)$$

Furthermore, considering the rotating Park reference frame such that the reference of the phases is taken according to the voltage of the first phase V_{san} , in this case, the quadrature component V_{sq} is zero where the direct component is constant and equal to $V_{sd} = |V_{san}|$. Under these conditions, the power expressions are reduced to:

$$P = \frac{3}{2} V_{sd} I_d \quad (9)$$

$$Q = \frac{3}{2} V_{sd} I_q \quad (10)$$

3. STATCOM control strategy

The STATCOM control system consists of two control loops. An external loop calculates the current reference to control the exchange of reactive power with the network. The internal current control loop calculates the reference voltages to generate the switch opening and closing commands using the adopted controller (Djellad et al., 2021).

3.1. STATCOM control using a conventional PI controller

Fig. 3 shows the schematic diagram of the STATCOM control system using a conventional PI controller (SPWM). The aim of the control system is to compensate the reactive power and to regulate voltage magnitude at the considered common connection point.

The control system receives the line-to-neutral voltages, the phase currents and the DC bus capacitor voltage as input signals. The PLL enables the controller to be synchronized with the phase of the signals. The quadrature component of the reference current I_{qref} is obtained by regulating the error between the reference and measured root-mean-square (RMS) values of the alternating voltage using separate PI controllers. Similarly, the error between the measured and reference values of the DC voltage is regulated through a separate PI controller to obtain the direct component of the reference current I_{dref} . The three-phase grid currents at the PCC are transformed into dq reference frame to create I_d and I_q . These currents are then compared to the corresponding reference values to create error signals (ΔI_d and ΔI_q) which are transmitted to the PI controller in order to compute the reference voltages components (V_{dref} and V_{qref}). Due to Park's inverse transformation, the voltages V_{dref} and V_{qref} are converted to V_{aref} , V_{bref} and V_{cref} . Finally, these voltages are compared with the triangular signals to generate the control signals (Murugesan and Muthu., 2011; Subsingha, 2016; Sadiq et al., 2021).

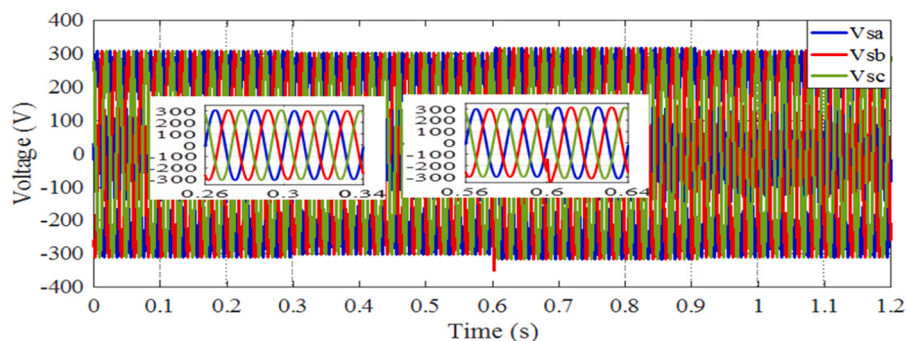


Fig. 9. Three-phase voltage (abc) of the source without STATCOM.

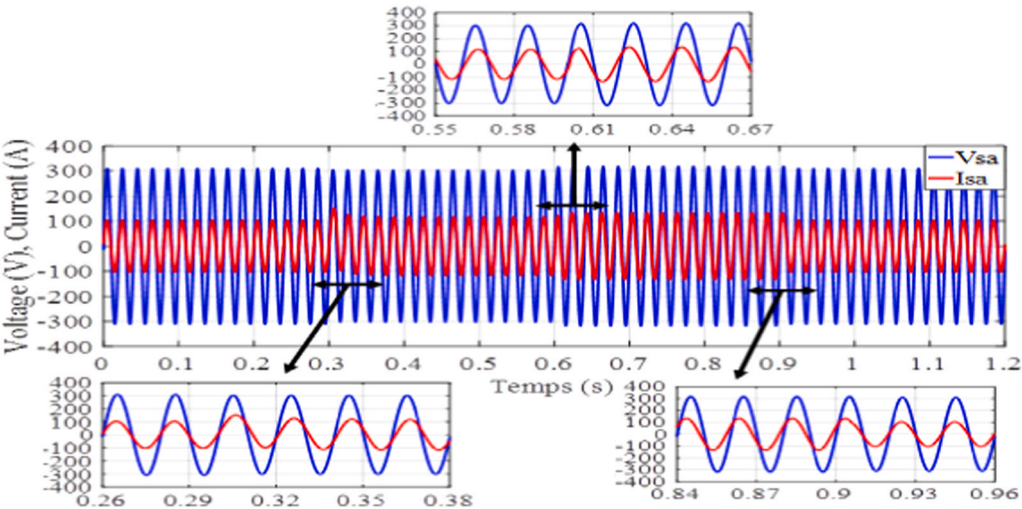


Fig. 10. Source voltage and current for phase (a) at PCC without STATCOM.

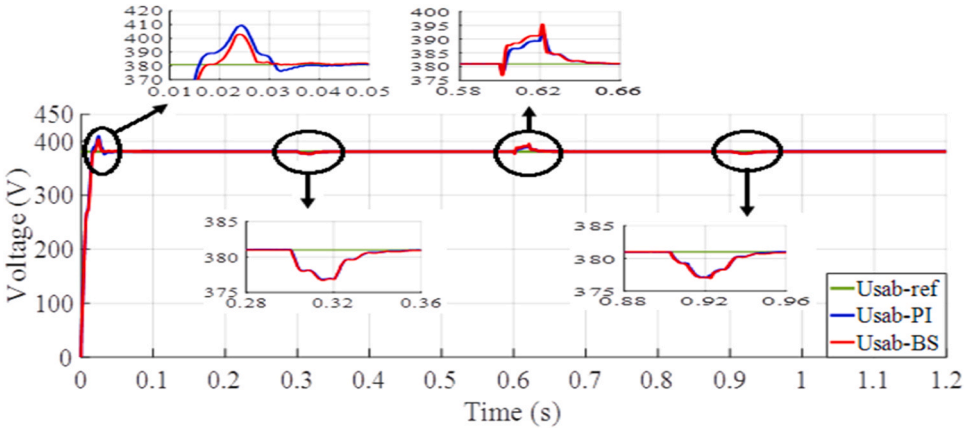


Fig. 11. RMS voltage between phases at PCC with STATCOM (backstepping-PI-control).

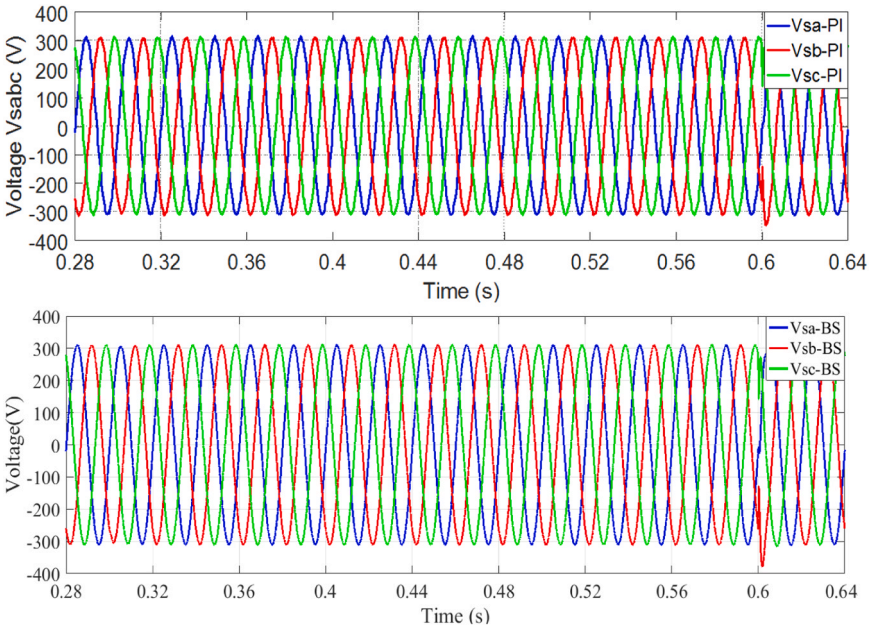


Fig. 12. Three-phase voltage (abc) of the source with STATCOM.

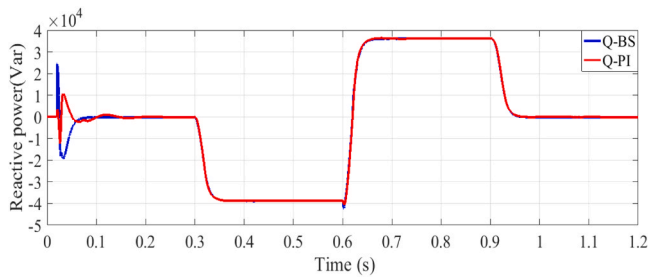


Fig. 13. Reactive power observed/supplied by STATCOM (backstepping-PI-control).

3.2. STATCOM control using a Backstepping controller (BSC) overview

The schematic diagram of the STATCOM control system using the Backstepping control approach is shown in Fig. 4.

Similar to the conventional PI control, the signals of the reference currents are calculated by the external control loop with the desired set value. The internal current control loop receives as an input signal the voltages V_{dq} , the currents I_{dq} and reference current I_{dqref} calculated by the external loop. The reference voltages V_{dref} and V_{qref} are performed by the

backstepping controller (BSC). Due to the inverse Park transformation, the voltages V_{dref} and V_{qref} are converted to V_{aref} , V_{bref} and V_{cref} . Finally, these voltages are compared with the triangular signals in order to generate the control signals.

3.2.1. Backstepping BSC procedure design for inductive and reactive currents regulations

The Backstepping technique is a relatively new control method for designing stabilizing control laws for the non-linear systems. It is based on the use of Lyapunov functions to linearize a non-linear system. The idea is to establish the control law of non-linear system considering certain state variables as being virtual controls and by designing intermediate control laws for them (El Ouanjli et al., 2019; Mazouz et al., 2021; Kokotović and Arcak, 2001).

The aim of the active current BSC design is to force the state variable of the active current (I_d) to follow the predefined reference I_{dref} . The error of the active current, i.e. (z_1), is defined as:

$$z_1 = I_{dref} - I_d \quad (11)$$

Its derivative is then defined by:

$$\dot{z}_1 = \dot{I}_{dref} - \dot{I}_d \quad (12)$$

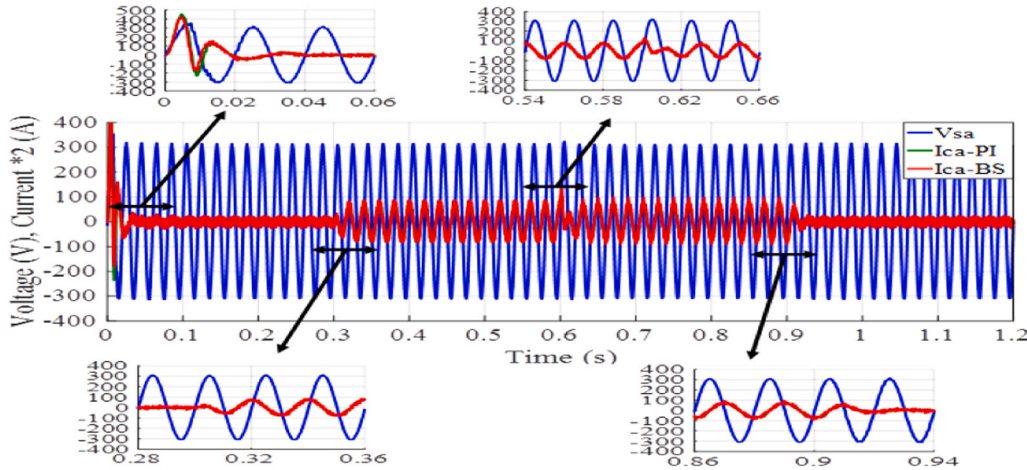


Fig. 14. Source voltage and current with STATCOM (backstepping-PI-control).

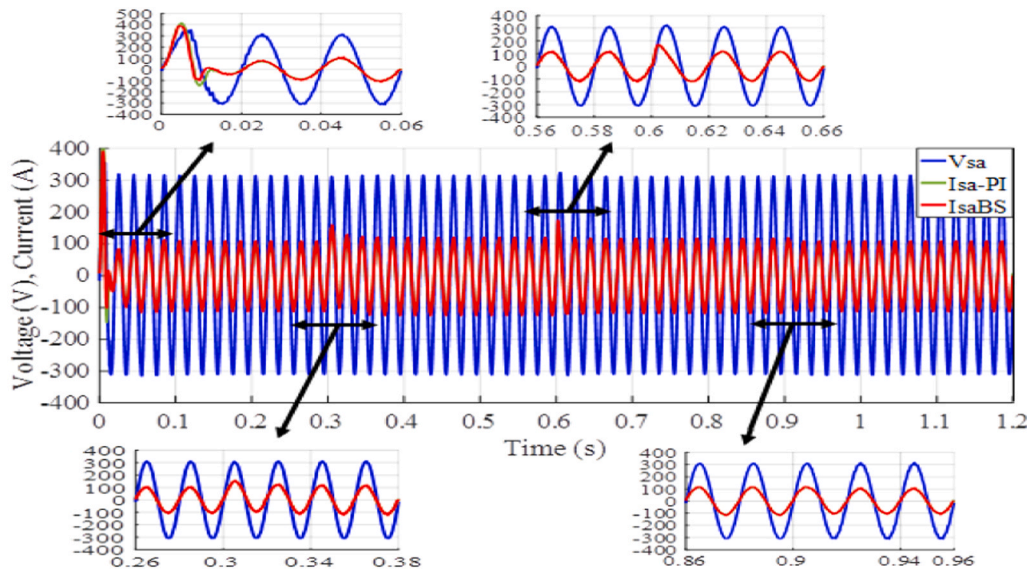


Fig. 15. Source voltage and current with STATCOM (backstepping-PI-control).

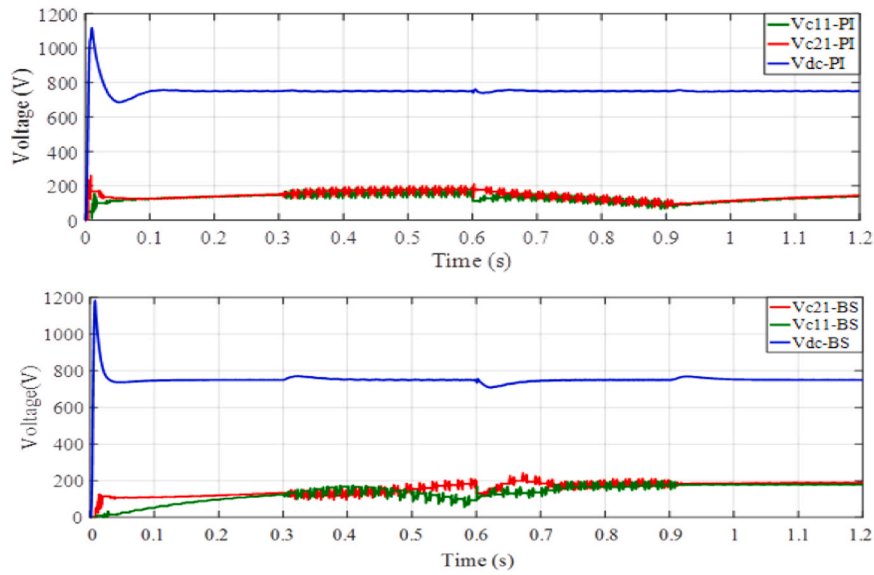


Fig. 16. DC voltage and source voltage curves with floating capacities.

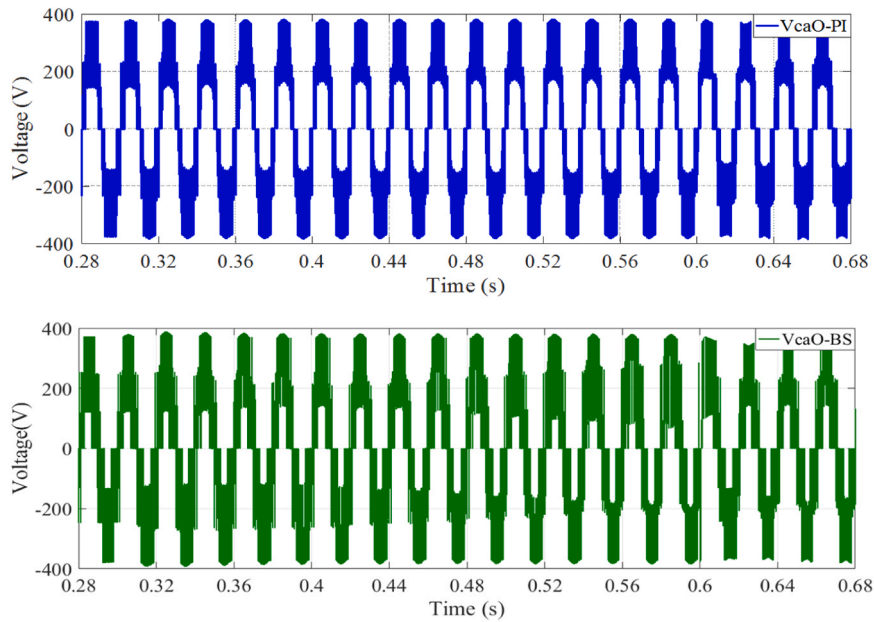


Fig. 17. Output voltage phase of the STATCOM converter (VcaO).

The candidate function of Lyapunov is defined by:

$$V_1 = \frac{1}{2}z_1^2 \quad (13)$$

By considering (6), the derivative of (13) is given by:

$$\dot{V}_1 = z_1 \dot{z}_1 = z_1 \left(\dot{I}_{dref} + \frac{r}{L}I_d + \omega I_q - \frac{V_{sd} - V_{dref}}{L} \right) \quad (14)$$

To ensure the stability of the Lyapunov function, its derivative must always be negative, under these conditions, the dynamic of Eq. 12 is chosen as (Kokotović and Arcak, 2001):

$$\dot{z}_1 = -k_1 z_1 \quad (15)$$

By replacing the expression of $\dot{z}_1$ in (13), we have:

$$\dot{I}_{dref} + \frac{R}{L}I_d + \omega I_q - \frac{\hat{V}_{sd} - V_{dref}}{L} = -k_1 z_1 \quad (16)$$

In this way, the control V_{dref} can be then defined by:

$$V_{dref} = L \left(-k_1 z_1 - \dot{I}_{dref} - \frac{R}{L}I_d - \omega I_q \right) + V_{sd} \quad (17)$$

The objective of the Backstepping control design of the reactive current is to force state variable of the active current (I_q) to follow the predefined reference I_{qref} . The error of the reactive current, i.e. (z_2), is defined as:

$$z_2 = I_{qref} - I_q \quad (18)$$

Its derivative is given by:

$$\dot{z}_2 = \dot{I}_{qref} - \dot{I}_q \quad (19)$$

$$\dot{z}_2 = \dot{I}_{qref} + \frac{R}{L}I_q - \omega I_d - \frac{V_{sq} - V_{qref}}{L} \quad (20)$$

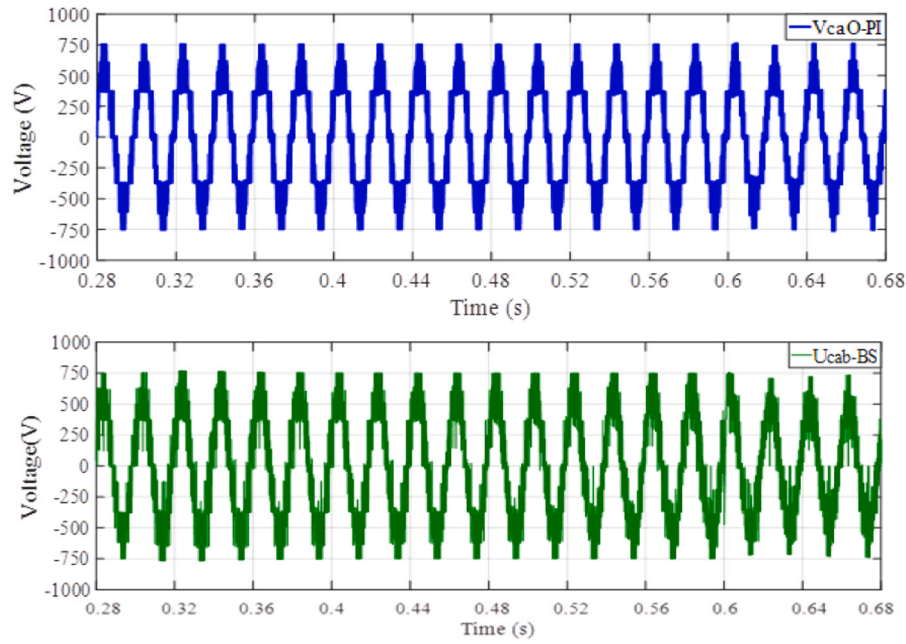


Fig. 18. Output voltage between phases of the STATCOM converter (Ucab).

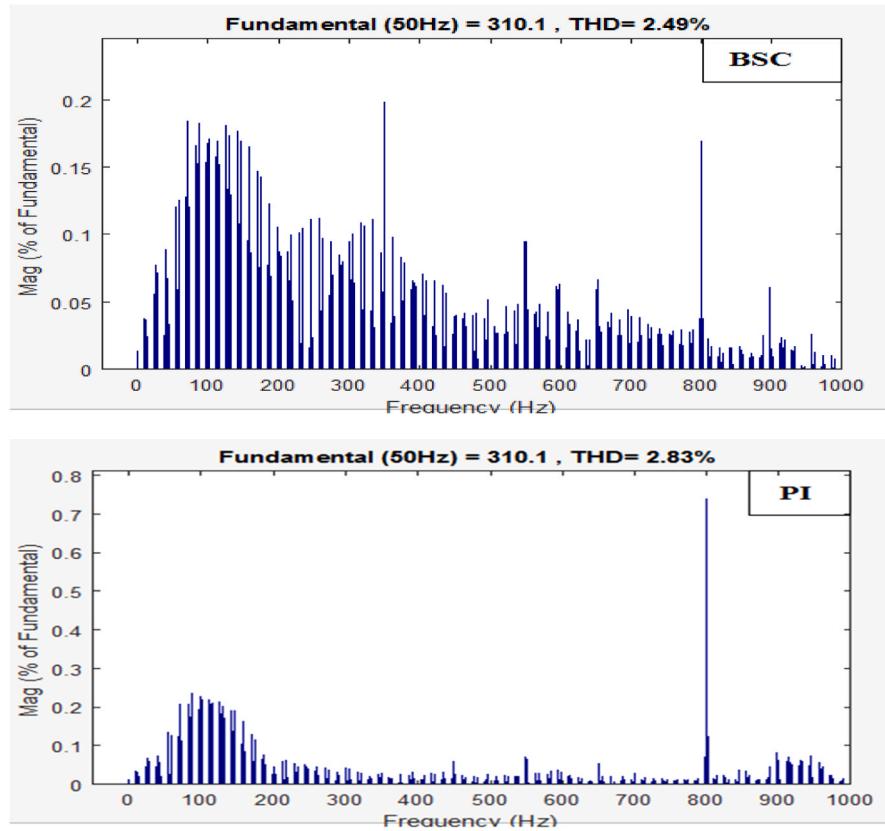


Fig. 19. Harmonic spectrum of phase voltage at PCC (Vsan).

Similarly, the corresponding Lyapunov function is chosen as:

$$V_2 = \frac{1}{2}z_2^2 \quad (21)$$

Applying the derivative to this function give:

$$\dot{V}_2 = z_2 \dot{z}_2 = z_2 \left(\dot{I}_{qref} + \frac{R}{L}I_q + \omega I_d - \frac{V_{sq} - V_{qref}}{L} \right) \quad (22)$$

In order to ensure the stability of the Lyapunov function, we choose:

$$\dot{z}_2 = -k_2 z_2 \quad (23)$$

Table 4
Comparative analysis.

	Voltage Vsan	
	THD (%)	FD (V)
BS Controller	2.49	310.1
PI Controller	2.83	310.1

Which gives:

$$\dot{I}_{qref} + \frac{R}{L}I_q - \omega I_d - \frac{V_{sq} - V_{qref}}{L} = -k_2 z_2 \quad (24)$$

Finally, the control V_{qref} is then defined by:

$$V_{qref} = L \left(-k_2 z_2 - \dot{I}_{qref} - \frac{R}{L}I_q + \omega I_d \right) + V_{sq} \quad (25)$$

3.3. PS-PWM strategy (Phase Shifted Pulse Width Modulation)

In this work, the adopted PWM technique for generating the control pulses of the superimposed multicell converter is that of the phase-shifted (PS-PWM). This technique is relatively simple to realize compared with the other techniques, and offers a low THD of the converter output voltage for all the modulation index values with a natural balancing for the voltage (Radan and Shahirinia, 2008; Renault et al., 2022).

The PS-PWM control of the converter is realized using a reference sinusoidal and two sets of carriers apart from each other by an angle of $2\pi/p$ (equal to π in this case). These carriers have the same frequency (f_p) and amplitude (A_p) as shown in Figs. 5 and 6. The upper carriers are generated between 0 and 1 are compared to the reference to generate the control signals of the cells of stage 1; when the lower carriers generated between -1 and 0 is compared to the reference, to generate the control signals of the cells of stage 2. Each comparison gives 1 if the reference is greater than or equal to a carrier, 0 in the opposite case (Pan et al., 2022).

4. Simulations and results analysis

To demonstrate the effectiveness of the proposed STATCOM solution relative to reactive energy compensation, the voltage regulation and the improvement of power factor at the considered point, the simulations were realized using a stacked superposed multicell converter five-level controlled by Backstepping control technique as well as the conventional PI-SVMPWM control according to Fig. 1. The whole system is simulated in MATLAB/Simulink by considering the system parameters shown in Table 3. The adopted simulation scenario is as follows: initially, the STATCOM compensator is not connected and the load (L1) is permanently applied. At time ($t = 0.3$ s), an inductive load (L2) is applied, and at time ($t = 0.6$ s), a capacitive load (L3) is applied while the inductive load (L2) is disconnected.

Consequently, the simulation shows both steady-state and transient operating zones. The sequence graph showing the scenario is defined in Fig. 7.

Fig. 8 show the RMS voltage between phases and the three-phase voltages at the point of common connection (PCC) without STATCOM in the presence of inductive and capacitive loads respectively. Inductive and capacitive loads destabilise the reactive power in the network resulting in a voltage dip of 5 % during the 0.3–0.6 s interval, then a voltage drop (overshoot) of around 5 % during the 0.6–0.9 s interval.

Fig. 9 shows the three-phase voltage waveform at PCC without STATCOM in the presence of inductive and capacitive loads respectively.

The voltage and current of the phase source is shown in Fig. 10. As it can be seen from this figure, the current lags the source voltage (inductive current) during the interval 0.3–0.6 s, then at the interval

0.6–0.9 s and leads the source voltage (capacitive current).

Fig. 11 shows the RMS voltage between phases at the common connection point (CCP) using STATCOM. As it is shown in this figure, STATCOM with the proposed control design can regulate the voltage at the PCC to the desired value (381 V), for both controllers. It deviates slightly from the reference value when inductive and capacitive loads are present. The deviations, due to the variation in reactive power that accompanies each load variation, nevertheless remain well below 4 %, and the response time is of the order of 0.04 s.

Fig. 12 illustrates the time profile of the three-phase voltages at the common connection point. It is clear in this figure that the voltage waveform at the PCC reaches the desired value for both controllers.

The STATCOM reacts positively and manages to keep the voltage at the PCC constant by controlling the amount of reactive power injected or absorbed into the network as shown in Fig. 13. In the range 0.3–0.6 s the STATCOM injects a quantity of reactive power to compensate the reactive power consumed by the inductive load. In this case, the compensator operates in capacitive mode. For the interval 0.6–0.9 s the STATCOM absorbs a quantity of reactive power to compensate for the reactive power supplied by the capacitive load; here, the compensator operates in inductive mode.

The corresponding exchange of reactive power takes place via the coupling leakage inductance, which helps to smooth the currents leading or lagging the angle with the voltages at the PCC, as shown in Fig. 14.

Fig. 15 shows the network voltage and current with STATCOM. It is confirmed that the power factor has increased up to 0.99, which verifies the satisfactory compensation effect of the proposed SMC-STATCOM compensator.

Fig. 16 illustrates the DC voltage and the internal floating capacitor voltages of the converter for phase a of STATCOM. It seems very clear that the DC voltage regulation block has proved its effectiveness in keeping the capacitor terminal voltage constant. It should be noted that the voltages of the internal floating capacitors are balanced at their desired voltages, which means that the natural self-balancing property is maintained.

Fig. 17 shows the voltage between phase a and point O (V_{ao}) at the converter output. As can be seen, the voltage levels are divided into five levels equal to -375 V, 0 V, 187.5 and 375 V.

Fig. 18 show the voltage between phase (U_{ab}), at the output of the Stacked Multi-cell converter of proposed STATCOM.

Fig. 19 shows the harmonic spectrum obtained from the phase voltage (Vsan) at the PCC using the FFT block for both controllers.

Table 4 shows total harmonic distortion of phase voltage (Van) at PCC for both controllers.

From Table 4, it is clear that the two structures give identical voltage values, and the Backstepping control technique gives better performance in terms of THD. However, the THD values of two techniques can be satisfied under IEEE Std. 519-1992.

5. Conclusion

This paper presents an analytical study of a STATCOM synchronous static compensator performances, based on a five-level stacked multicell converter controlled by two control techniques Backstepping and conventional PI. The mathematical model of the proposed STATCOM was detailed in order to perform correctly the adopted Backstepping controller design. The structure of the proposed STATCOM was simulated in the Simulink/Matlab environment. The obtained results proved that this structure containing two current controllers is plausible and provide perfect voltage quantities (DC and AC side) tracking performances with a response time of around 0.05 s whatever the type of disturbance (dip or overvoltage) with a good THD factor. Furthermore, a unity power factor on the grid side was achieved independently of the considered load variations. The simplicity of the backstepping control law developed during this work makes this control strategy very useful for possible real-time implementation.

CRediT authorship contribution statement

Mohamed Lamine Srief: Writing – review & editing, Writing – original draft, Software, Formal analysis. **Ghennai Malak:** Formal analysis. **Nemmour Ahmed Lokmane:** Validation. **Belakehal Soltane:** Visualization, Validation, Supervision.

Author Statement

I am an author on this submission, have adhered to all editorial policies for submission as described in the Information for Authors, attest to having met all authorship criteria, and all potential conflicts of interest/financial disclosures appears on the title page of the submission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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